

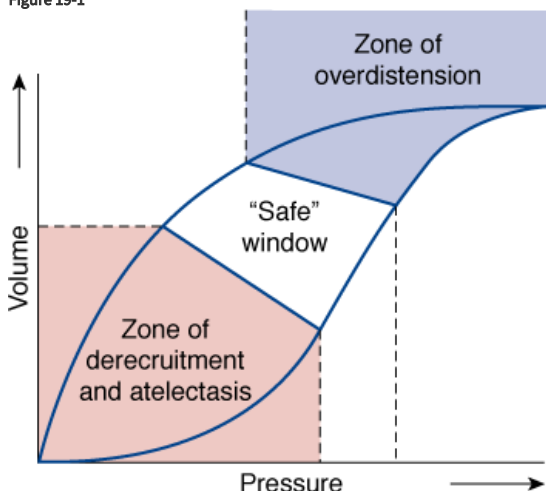
Chapter 19. High-Frequency Ventilation

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High-Frequency Ventilation: Introduction

High-frequency ventilation has been an unconventional option for more than three decades and during that period several varieties of high-frequency ventilators have come and gone. Currently, interest in high-frequency ventilation in adult critical care is part of a larger search for ventilator strategies that can support gas exchange in the severely hypoxemic patient without contributing additional ventilator-induced lung injury. Over the past 30 years, high-frequency ventilators provided an experimental tool that identified many of the mechanisms that contribute to ventilator-induced lung injury. It became clear that ventilator-induced lung injury is minimized by ventilator patterns that achieve homogeneous aeration of as much of the lung as possible, avoiding both injury from overdistension (volutrauma) and that arising from the repetitive opening and closing of lung units in regions of atelectasis (atelectrauma) (Fig. 19-1).¹ Failure to operate in the “safe zone” initiates biotrauma.^{2,3}

Figure 19-1



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Pressure-volume curve of a moderately diseased lung, as in a patient with acute lung injury. Ventilator-induced lung injury occurs at both extremes of lung volume. In the zone of overdistension, damage arises from edema fluid accumulation, surfactant degradation, and mechanical disruption. In the zone of derecruitment and atelectasis, lung injury arises from the direct trauma of repeated closure and reexpansion of small airways and alveoli, through accumulation and activation of inflammatory cells with release of cytokines (biotrauma), through interactions with local hypoxemia, by inhibition of surfactant, and through compensatory overexpansion of the rest of the lung as the lung “shrinks.” High end-expiratory pressures plus small tidal volume cycles are needed to stay in the safe window. (Reproduced, with permission, from Lippincott Williams & Wilkins, Froese AB, High-frequency oscillatory ventilation for adult respiratory distress syndrome: let’s get it right this time. *Crit Care Med*. 1997;25:906–908.)

The concept of a “safe zone” within which to ventilate the atelectasis-prone lung has been reflected in numerous clinical trials of lung-protective ventilation over the last 15 years. Conventional ventilator protocols have found survival benefit from shrinking the tidal volume and minimizing peak or plateau distending pressures.⁴ Studies such as that of Roupie et al⁵ and Terragni et al⁶ suggest that very small tidal volumes—even lower than 6 mL/kg predicted body weight—may be needed in some patients (those with the worst lung injury) to avoid overdistension. Very high levels of positive end-expiratory pressure (PEEP) would be needed in some patients to avoid derecruitment.⁷ Concurrently, high-frequency ventilation—both in oscillatory and jet forms—has become an established lung-protective modality in neonatal and pediatric intensive care (see Chapter 23).^{8–14} The question, however, persists: in severe acute respiratory distress syndrome in adult

patients, will use of a high-frequency device result in clinically important outcome differences compared with lung-protective conventional ventilation, or are newer conventional ventilator protocols now equally able to protect the lung?^{15–17}

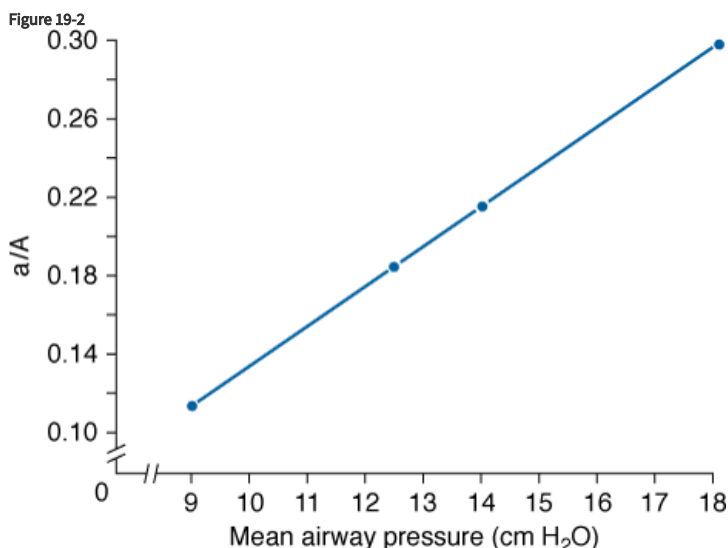
Historical Overview

Several existing reviews detail the history of high-frequency ventilation.^{18,19} Early developments often were driven by issues peripheral to pulmonary critical care. Sjöstrand et al²⁰ wanted to eliminate respiration-related variations in vascular pressures so that they could investigate carotid sinus reflexes and developed high-frequency positive-pressure ventilation. Lunkenheimer et al²¹ wanted to use the lungs as a route to deliver oscillatory pressure pulses to the myocardium. They needed apnea for this and were amazed to find gas exchange occurring while they applied their high-frequency flow oscillations. Emerson²² thought that high-frequency flow oscillations might provide internal physiotherapy and help to mobilize secretions. In Toronto, Bryan²³ initially was curious to see whether an external “shaker” could enhance the gas mixing produced by cardiogenic oscillations. Klain and Smith²⁴ explored jet ventilation at increasing frequencies to solve the problem of achieving alveolar ventilation in respiratory systems with a big leak, such as a bronchopleural fistula.

These devices often became intriguing phenomena in search of a reason for being. Devices such as high-frequency positive-pressure ventilation and high-frequency jet ventilation were particularly useful for surgical procedures when both anesthesiologist and surgeon needed access to the airway. The notion, however, that high rates and small tidal volumes might be of broader therapeutic value needed a pathophysiologic rationale.

An emerging concept in the 1970s was that many of the pulmonary perturbations that put patients into intensive care units were problems of low lung volume. Low lung volumes were associated with poor lung compliance, increasing airway and vascular resistances, increased work of breathing, airway closure, low ventilation–perfusion ($\dot{V}_A/\dot{Q}$) ratios, atelectasis, hypoxemia, and high oxygen (O_2) exposure. Neonatal outcome was improving simply from the introduction of continuous positive airway pressure to improve alveolar aeration.²⁵

A crucial insight occurred early in our experience with high-frequency oscillatory ventilation (HFOV) when we explored a variety of mean airway pressure settings while ventilating an infant with neonatal respiratory distress syndrome (Fig. 19-2).²⁶ The infant was stable in terms of both hemodynamics and CO_2 elimination over the whole range of mean pressures tested, but oxygenation varied markedly. One could either ventilate with a low mean airway pressure (mPaw) and high fractional inspired oxygen concentration (FI_{O_2}) or a higher mean pressure and low FI_{O_2} . A choice had to be made. We gave priority to the reversal of low lung volumes. We argued that the small-volume cycles of HFOV should allow one to optimize alveolar aeration by using higher mean pressures than were considered safe during conventional mechanical ventilation while still keeping peak pressures less than those needed to eliminate CO_2 at conventional rates. This pathophysiologic rationale continues to guide high-frequency applications to this day.



Relationship of oxygenation, as reflected in the ratio of the arterial and alveolar **oxygen** tensions (a/A ratio), to the mean airway pressure (mPaw) applied during HFOV, at constant tidal volume and frequency, in an infant with respiratory distress syndrome. No circulatory instability occurred over the entire range of mean pressures. CO₂ elimination could be achieved equally well using a high F_IO₂ and a low mPaw or a low F_IO₂ and a higher mPaw. A choice of operating conditions had to be made. (From Marchak, et al. Treatment of RDS by high-frequency oscillatory ventilation: a preliminary report. *J Pediatr*. 1981;99:287–292. Reproduced with permission of Mosby, Inc.)

Many devices invented along the way, such as high-frequency positive-pressure ventilation, have since disappeared from use. High-frequency jet ventilation was explored in many adult intensive care units in the 1980s for difficult cases of bronchopleural fistula or tracheal disruption.²⁷ It gradually became clear that any oxygenation benefits occurring during high-frequency jet ventilation resulted from increases in mean lung volume, not from some unusual properties of gas distribution.²⁸ With an high-frequency jet ventilation device, the end-expiratory lung volume is a complex product of jet diameter and placement, driving pressure, jet frequency, and the time available for expiration.^{29,30} Safe use requires accurate intrapulmonary pressure monitoring, which was rarely provided with early devices. Inadvertent hyperinflation could cause problems with both circulatory depression and/or barotrauma. No North American, commercial, adult-sized jet ventilator was ever marketed with a safe, effective humidification system such as that available in Europe.

The largest early comparative trial of high-frequency jet ventilation versus conventional mechanical ventilation in hypoxemic lung disease was performed before the importance of atelectasis reversal was established. Consequently, it was carried out with the goal of supporting gas exchange with the lowest possible peak and mean pressures, and proved of no benefit.³¹ For all these reasons, high-frequency jet ventilation in adults has become a rare event. Jet ventilation at high or low frequencies continues to be a useful approach to situations in which airway access must be shared during surgical procedures³² or alveolar ventilation needs to be maintained in the presence of severe bronchopleural fistulas.²⁹

High-frequency jet ventilation has persisted in neonatal and pediatric critical care largely because of the continuous design refinements of the Bunnell Life Pulse device.^{13,14} Early problems with poor humidification causing desiccation of tracheal mucosa were corrected, appropriate pressure-monitoring systems were devised, and expert training and technical support were provided.

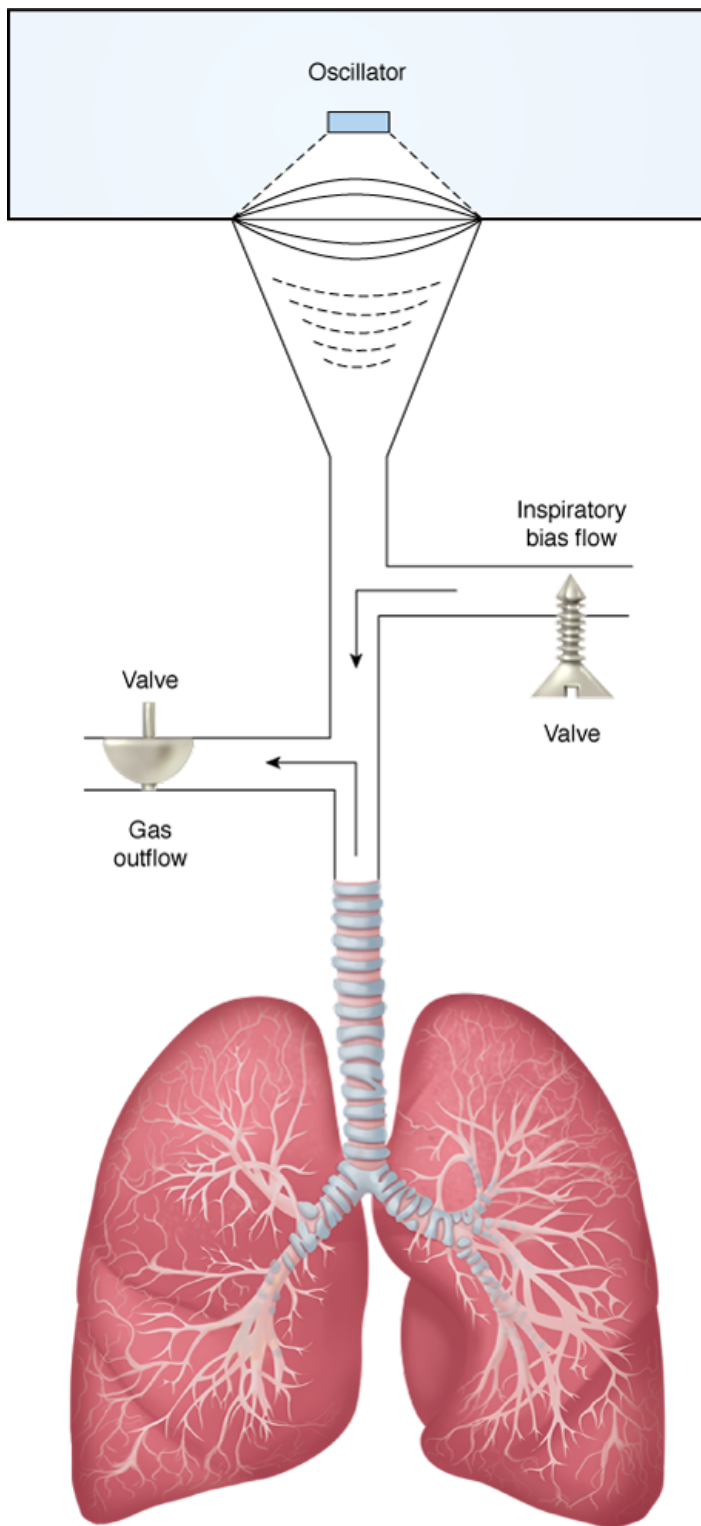
A hybrid device combining high-frequency ventilation and conventional pressure-cycled ventilation (e.g., the high-frequency percussive ventilation/volume diffusive respirator [Percussionaire 4, Bird Technologies, Sandpoint, ID]) is used in many burn units.³³ Its use was reviewed recently.³⁴ Currently, the main high-frequency ventilatory options in the critical care of adults or larger children are HFOV or the high-frequency percussive ventilation/volume diffusive respirator Percussionaire. For neonates and small infants, high-frequency oscillatory ventilators, high-frequency jet ventilators, and the high-frequency percussive ventilator remain available. In view of HFOV's increasing role in adult intensive care, this chapter focuses on HFOV.

Basic Principles of High-Frequency Oscillatory Ventilation

HFOV achieves gas transport with stroke volumes approximating anatomic dead space. Quasi-sinusoidal flow oscillations applied at the airway opening induce rapid gas mixing within the lungs. A number of physical transport mechanisms contribute to this mixing process. They have been reviewed in detail elsewhere.³⁵

In practical terms, HFOV can be viewed as a mixing device that rapidly blends high O₂/low CO₂ gas from the top of the endotracheal tube with gas in the alveoli (Fig. 19-3). Net transport occurs along the partial-pressure gradients for O₂ and CO₂, with CO₂ moving out of the lung along its partial-pressure gradient and O₂ moving inward to the alveolar–capillary interface. These flow oscillations cause symmetric oscillations of intrapulmonary pressure (ΔP) around a mean distending airway pressure (mPaw) (Fig. 19-4). Although subambient pressures can occur in the circuit, intrapulmonary air trapping or “choke points” are unlikely with appropriate mPaw settings.^{36,37} One can view HFOV as a means of delivering “continuous positive airway pressure” with a built-in “shaker” to facilitate CO₂ elimination.

Figure 19-3



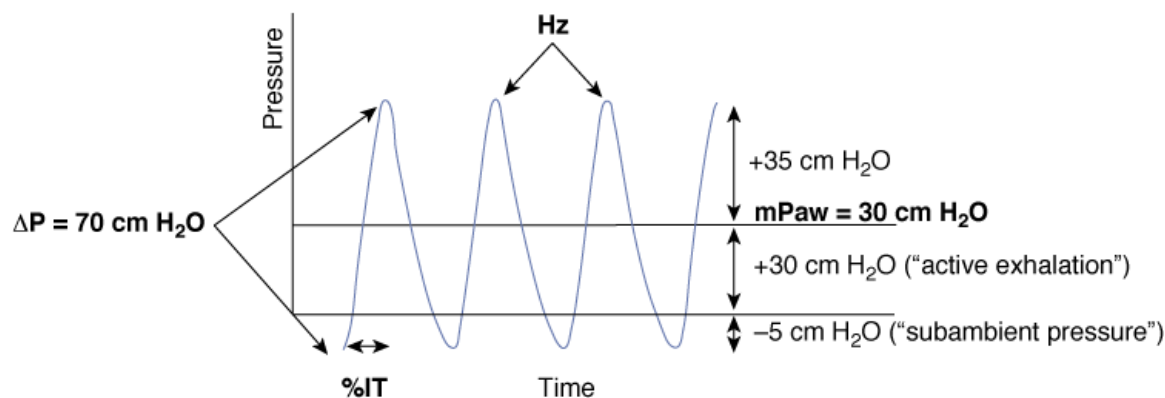
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Schematic of a circuit for delivery of HFOV. Quasi-sinusoidal flow oscillations are generated by a diaphragm or piston pump and directed to the endotracheal tube. A bias flow of humidified gas flushes the CO₂ that is transported out of the lungs out of the circuit. Mean airway pressure is regulated by adjustments of the bias flow and the resistance of the expiratory limb. (From Krishnan, et al. High-frequency ventilation for acute lung injury and ARDS. *Chest*. 2000;118:795–807. Reproduced with the permission of the American College of Chest Physicians.)

Figure 19-4

HFOV pressure waveform



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Waveform of high-frequency pressure oscillations in the circuit above the endotracheal tube. Both inspiratory and expiratory flows are actively driven by the oscillator diaphragm. Pressure cycles equally above and below the mean level. When the oscillatory pressure amplitude (ΔP) is more than twice the mean airway pressure (mPaw), subambient pressures may occur in the circuit without inducing air trapping or choke points in the lung.³⁷ The endotracheal tube filters the pressure swings, decreasing ΔP in the trachea and alveoli. Filtering is greater with smaller endotracheal tubes and higher frequencies.^{71,84,96} (From Derdak S. High-frequency oscillatory ventilation for acute respiratory distress syndrome in adult patients. *Crit Care Med*. 2003;31:S317–S323. Reproduced with permission from Lippincott, Williams & Wilkins.)

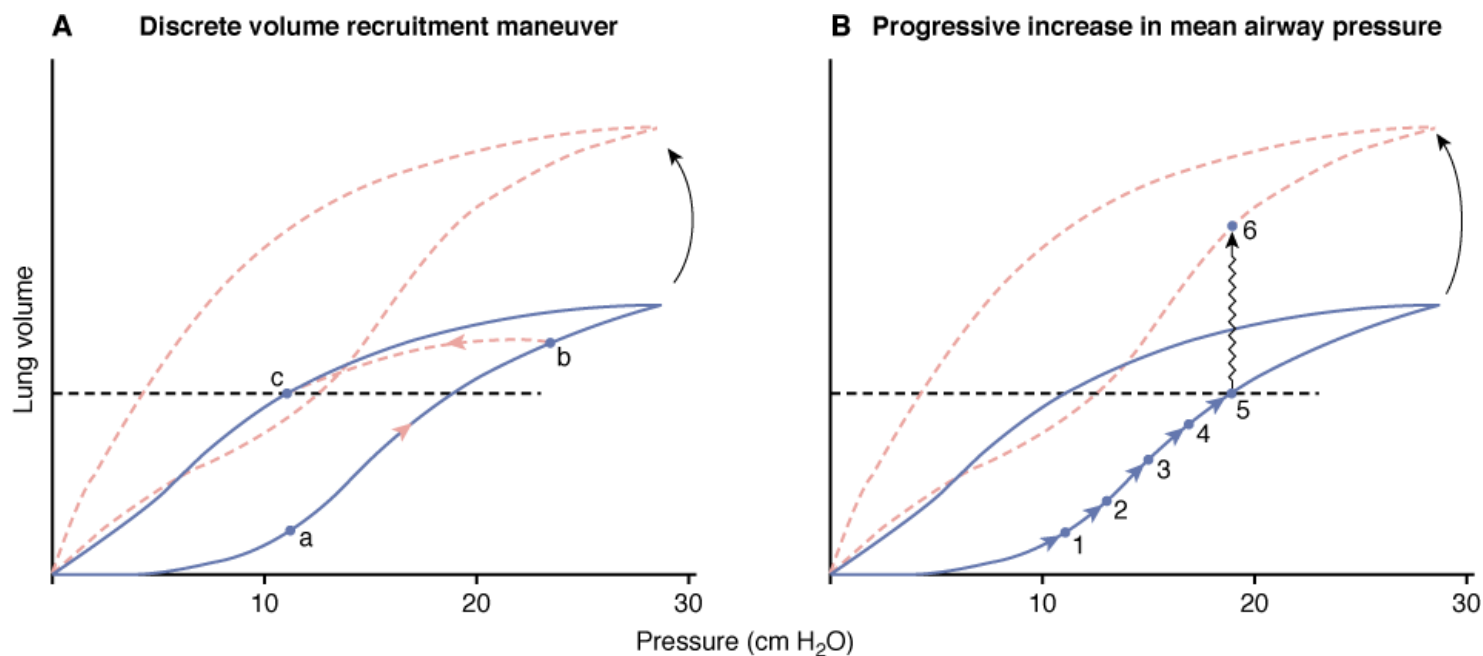
In contrast to current conventional approaches to supplying PEEP with additional inspiratory assist for CO₂ elimination, the mean distending pressure during HFOV is midway between the minimal and maximum values, introducing less risk of derecruitment during the expiratory phase for any given peak distending volume or pressure.³⁸ One attraction of HFOV has been the way in which it uncouples the regulation of oxygenation and CO₂ elimination into two separate control systems, unlike the situation with conventional ventilators, where it is often difficult to adjust one (i.e., the CO₂ level) without also affecting the other. Oxygenation is regulated by reversing atelectasis and then finding the mean distending pressure that maintains alveolar expansion. The FiO₂ then is set at a level that maintains appropriate arterial oxygenation goals. CO₂ elimination is relatively independent of mean airway pressure,³⁹ being regulated by frequency and stroke volume (i.e., power or ΔP) adjustments.^{40,41}

Oxygenation

Achieving Alveolar Aeration

Although lung-volume optimization has become an accepted goal of HFOV, the “best way” to achieve this optimization remains controversial (Fig. 19-5). The small volume cycles of HFOV are simply not powerful enough to reopen atelectatic alveoli rapidly without some type of recruitment measure.

Figure 19-5



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Schematic of two approaches to alveolar reexpansion during HFOV. The *horizontal dashed line* indicates the desired maintenance mean lung volume. The *solid line* is the pressure–volume relationship of a moderately diseased lung that still exhibits hysteresis. The *dashed line* indicates the pressure–volume relationship after some recovery has occurred. **A.** A brief sustained increase in mean airway pressure (mPaw) from *a* to *b* inflates the lung to near total lung capacity, putting it on the deflation limb of its pressure–volume curve. After this discrete recruitment maneuver, the target volume *c* is maintained at an mPaw of 11 cm H₂O. If the pressure–volume relationship happens to change (as with position, diuresis, etc.), the operational lung volume remains constant. This is the approach used in the Treatment with Oscillation and an Open Lung Strategy (TOOLS) trial.⁵⁴ **B.** A gradual march up the inflation limb of the pressure–volume relationship. Progressive increases in mPaw are used to achieve the target lung volume. An mPaw of 19 cm H₂O is now needed to maintain lung volume at *c*. Also, if the pressure–volume relationship of the respiratory system changes, overdistension of the lung could occur (i.e., movement from point 5 to point 6). This is the approach used in the majority of neonatal and adult trials of HFOV to date. When actual lung volumes are measured, settings thought to have optimized lung volume clinically by these stepwise increases in mPaw often prove to be inadequate.⁶¹ (Froese AB. Neonatal and pediatric ventilation: Physiological and clinical perspectives. In: Marini JJ, Slutsky AS, eds. *Physiological Basis of Ventilatory Support*. 1998:1315–1357.)

Ventilating on the “Deflation Limb”

All initial animal studies and one early human trial used a brief recruitment maneuver to near total lung capacity, followed by reduction of mPaw to a maintenance level that prevented derecruitment.^{42–46} This is termed getting the lung *onto the deflation limb* of its pressure–volume relationship. Recent investigations in both animal models and humans reinforce the value of ventilating on the deflation limb.^{47–54} After a recruitment maneuver, oxygenation goals are achieved at substantially lower maintenance mPaw values (near the point of maximum curvature),^{48,51,52} alveolar expansion becomes more homogeneous (which should reduce shear forces during ventilation and reduce volutrauma),^{50,55} and the percentage transmission of HFOV pressure cycles into the lung is decreased relative to settings producing equal shunt reduction on the inflation limb.⁴⁹ Most studies in animals and humans have used one or more sustained inflation recruitment maneuvers to get the lung onto the deflation limb. A recent study in saline-lavaged pigs concluded that an escalating stepwise slow recruitment maneuver was more effective in opening the lung than brief sighs or a 20-second sustained inflation.⁵⁶ This study, however, is limited by the low pressures employed and the incomplete recruitment seen with all techniques tested.

Ventilating on the “Inflation Limb”

Following the High-Frequency Intervention (HIFI) trial of the late 1980s,⁵⁷ fear of intraventricular hemorrhage in the fragile brain of the premature patient led most neonatologists to pursue stepwise increases in mPaw until either X-ray and blood-gas evidence of lung

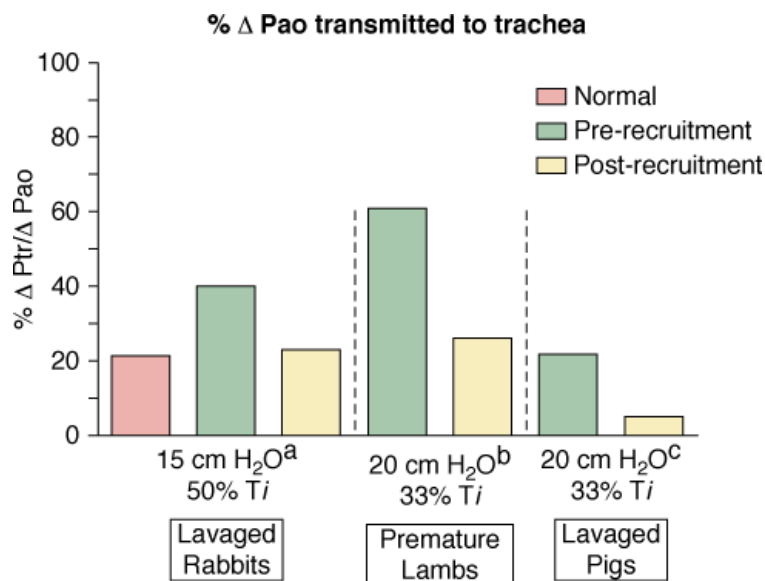
reexpansion was achieved or the mPaw level reached whatever level was deemed the “maximum allowable” level for a given institution.⁵⁸ Physiologically, this can be described as “marching up the inflation limb” of the pressure–volume curve, as in [Figure 19-5B](#). Most post-HIFI trial neonatal/pediatric trials (including the largest recent randomized, controlled trials in 2002^{59,60}) used stepwise increases in mPaw to “optimize” lung volume to oxygenation and chest X-ray targets. Unfortunately, when the actual lung volumes achieved by such clinical protocols are measured, many lungs prove to be suboptimally expanded.⁶¹ It follows that many comparative trials of HFOV and lung-protective conventional ventilation over the last 25 years have not achieved optimal homogeneous alveolar aeration at the lowest effective mean airway pressure, thus missing some of the potentially protective features of HFOV. A recent analysis of neonatal trials identified many design problems and stressed the need for further well-designed and monitored comparisons of ventilator options in this population.⁶² All early adult trials of HFOV started at mPaw levels of 3 to 5 cm H₂O above the mPaw on conventional ventilation and then increased mPaw incrementally—again marching up the inflation limb.^{63–67}

A few human trials have used protocols that rapidly place the lung “on the deflation limb” of its pressure–volume relationship.^{53,54} Although large comparative trials of the relative safety of these different recruitment protocols are not available, no evidence of risk from either barotrauma or intraventricular hemorrhage (IVH) has emerged using any of the neonatal or pediatric approaches since optimization of lung volume became an accepted goal of HFOV.^{8,9} What we do know is that lung-volume optimization is essential for HFOV to be protective of the lung, and alveolar aeration is more homogeneous and achieved at a lower mPaw after a recruitment maneuver, with or without use of the prone position, has moved the lung onto the deflation limb of the pressure–volume relationship. In a recent human trial, the oxygenation improvements gained in the prone position were only sustained when patients were managed with HFOV after being placed supine again.⁶⁸ These observations suggest that the higher mPaw and small pressure and volume cycles of HFOV preserve alveolar reexpansion better than the current lung-protective tidal volumes of conventional ventilation.

Atelectrauma: The Costs of Inadequate Recruitment

The relationship between the pressure swings at the airway opening (ΔP_{ao}) and those applied to the airways and alveoli ($\Delta P_{tr}/\Delta P_{ao}$) is multifactorial. Atelectasis markedly impacts the percentage pressure transmission.^{69–72} Premature lambs,⁶⁹ lavaged pigs,⁴⁹ and lavaged rabbits⁷² all demonstrate the high cost of failure to reverse atelectasis, as reflected in the percentage of ΔP_{ao} transmitted to the trachea, bronchi, and alveoli ([Fig. 19-6](#)). Pressure swings and the risk of overdistension of aerated lung reduce rapidly to normal values following lung recruitment. Failure to reverse atelectasis during HFOV exposes airways and alveoli unnecessarily to potentially damaging shear forces. When van Genderingen et al⁴⁹ analyzed intrapulmonary pressure transmission in terms of the ratio of intratracheal and circuit (machine displayed) ΔP values, termed the *oscillatory pressure ratio*, pressure transmission into the lung increased substantially at low lung volume. In vivo, adequate alveolar reexpansion optimized oxygenation while also minimizing intrapulmonary pressure swings. For a given level of shunt reduction, both the oscillatory pressure ratio and the mPaw needed to achieve optimal gas exchange were substantially lower after a recruitment maneuver.

Figure 19-6



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Effect of a recruitment maneuver (RM) on the transmission of ΔP from the oscillator circuit (ΔP_{ao}) to a tracheal sampling site (ΔP_{tr}). Alveolar reexpansion from an RM substantially reduced the amplitude of the peak-to-trough pressure swing applied to the trachea in lavaged rabbits,⁷² premature lambs,⁶⁹ and lavaged pigs.⁴⁹ Reversal of atelectasis protects lung tissue from potentially damaging shear forces. Note that $\Delta P_{machine} = \Delta P_{circuit} = \Delta P_{ao}$ in various sources.

Inadequate alveolar recruitment also may trigger inappropriate increases in ΔP or decreases in frequency because of hypercapnia, when what was really needed was alveolar reexpansion. All high-frequency oscillators currently available are load-dependent and deliver less tidal volume at any given setting when the lung is stiffer, as with ongoing atelectasis.^{73,74} If one's goal is to explore HFOV for the potential lung-protective effects of maintaining O₂ and CO₂ transport at the smallest possible intrapulmonary pressure and volume swings, then lung-volume optimization is essential.

Volutrauma: The Cost of Excessive Recruitment

In the presence of recruitable lung, it costs less to pursue lung recruitment vigorously than to accept ongoing atelectasis.⁷⁵ In late-phase disease with bronchiectasis, cysts, and progressive fibrosis, the opportunity for effective recruitment is past and should not be pursued. Similarly, in patients with milder acute lung injury (ratio of arterial partial pressure of oxygen to F_{IO₂} is 200 to 300) whose lungs are already reasonably well aerated, attempts at further lung recruitment will likely only result in overdistension of the already open lung.⁷⁶ All clinical trials done to date—from neonate to adult—that have mandated earlier lung-volume optimization have reported similar or decreased incidences of barotrauma, a transient need for inotropic support, and a decreased need for nitric oxide to reduce pulmonary vascular resistance despite using mean pressures in the thirties or forties when necessary to achieve oxygenation.^{11,64,77-80} Recent computed tomography studies during HFOV in eight adult patients with acute respiratory distress syndrome documented substantial (approximately 800 mL) increases in normally aerated volume with only a minor increase in hyperinflated lung (<50 mL).⁸¹ Overdistension certainly can occur, particularly if inappropriately high mPaw levels are maintained as the lung normalizes during treatment. Avoidance of this requires awareness and periodic reassessment of lung expansion.

Some algorithms warn against using mean distending pressures on HFOV higher than the plateau pressure limit of 30 cm H₂O currently advocated for conventional ventilation. This warning may produce inadequate volume optimization, particularly in extrapulmonary acute respiratory distress syndrome. The risk of a given “maximum pressure” will vary with the size of the lung and the size and rate of delivery of the tidal volume engendering that plateau pressure. If 60% or 70% of the lung is not participating in gas exchange, then a tidal volume of even 6 mL/kg may induce dangerous overdistension of the ventilated alveolar units, as well as exerting shear forces on any alveolar units reexpanded during the course of the breath. If, instead, 70% or 80% of the lung is kept expanded throughout the ventilatory cycle, then a mean distending pressure of 30 cm H₂O or more is safe, as proven in numerous clinical trials of HFOV,^{65,77,78,82} presumably because that distending pressure is now accommodated within many more participating lung units, fewer units are sheared open during the course of a cycle,³⁸ and volume

swings of only 1.5 to 2 mL/kg at high rates minimize peak distension. Conversely, if one fails to open the lung during HFOV and then keep it open, one will expose the lung to unwanted shear forces many more times per minute, particularly if forced to use lower rates and larger volume cycles because of inadequate recruitment.

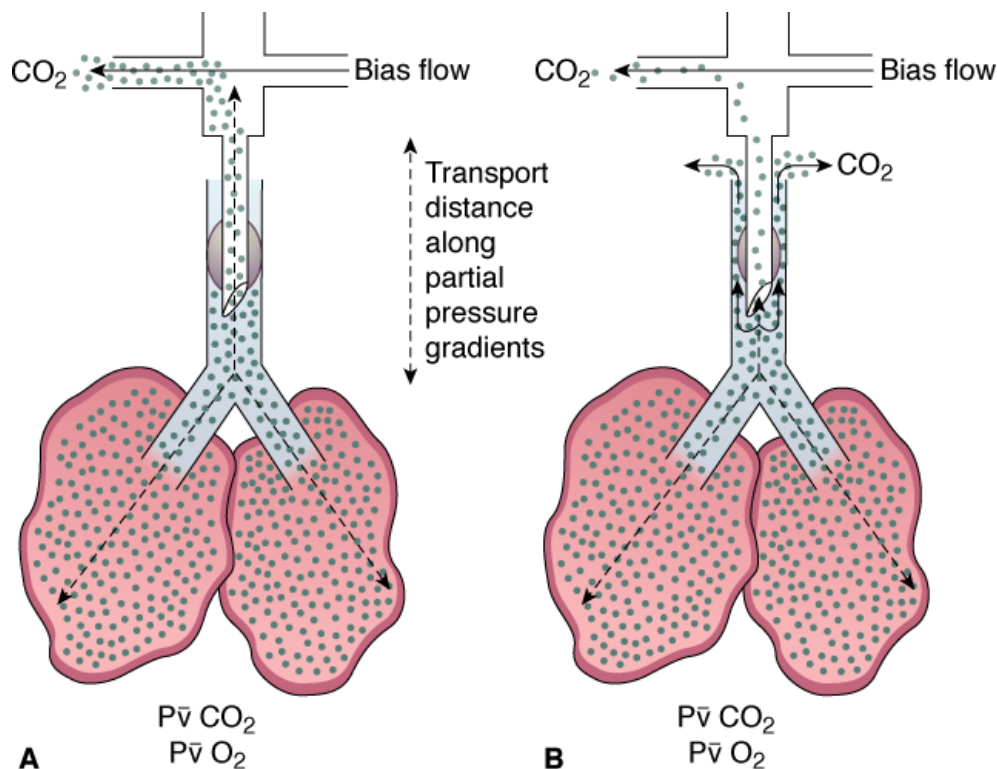
Maintenance Mean Airway Pressure

Numerous studies have tried to predict an optimal maintenance mPaw from some feature of the static pressure–volume curve of the respiratory system, such as the inflection point on the inflation limb.⁵¹ These studies demonstrate a general property of the respiratory system, namely, that the mPaw needed to maintain a given level of oxygenation is substantially less after the lung has been inflated briefly to a pressure of 30 to 40 cm H₂O so that it is “operating on the deflation limb.”⁴⁹ Because recruitment and derecruitment pressures vary markedly between lung regions^{50,83} and between lung-injury models, generalizations to human ventilator management must be made cautiously. In clinical practice, a gradual deterioration in oxygenation over time that corrects with a recruitment maneuver indicates that the mPaw needs to be increased to keep alveoli/airways above their closing pressures. Conversely, a rise in the partial pressure of arterial carbon dioxide (PaCO₂) at constant mPaw and power settings may indicate the lung is improving and is now overdistended, in which case a decrease in mPaw is needed.⁸⁴ Nothing can replace thoughtful reassessment of both patient and machine factors.⁸⁵ New bedside regional lung-volume monitoring techniques, such as electrical impedance tomography or respiratory inductance plethysmography, have the potential to clarify mPaw adjustments and may become more commonly used in the future.^{48,86–88}

Carbon Dioxide Elimination

The basic transport studies of the 1980s were executed with accurate measurements of delivered stroke volumes over a wide range of frequencies using sinusoidal flow oscillations.^{40,89,90} They established that the volume of CO₂ eliminated per minute ($\dot{V}_{\text{CO}_2}$) was proportional to the product of oscillatory frequency and the approximate square of the stroke volume ($\dot{V}_{\text{CO}_2} = V_T^2 \cdot f$). The endotracheal tube contributes a substantial impedance to gas transport during HFOV.⁹¹ If one uses an uncuffed endotracheal tube, or employs a partial cuff leak around a cuffed tube, such that the fresh-gas front moves to the bottom of the tube and CO₂ exits around rather than through the tube, the stroke volume needed to achieve normocapnia decreases by up to 50% (Fig. 19-7). The bias gas-flow rate also influences transport efficiency through its effect on gas tensions at the fresh-gas front.⁹² Uncuffed tubes are used routinely in neonates and infants. Cuff leaks substantially assist CO₂ elimination in adults,^{41,66} at the same time reducing intrapulmonary pressure swings.⁸⁴

Figure 19-7



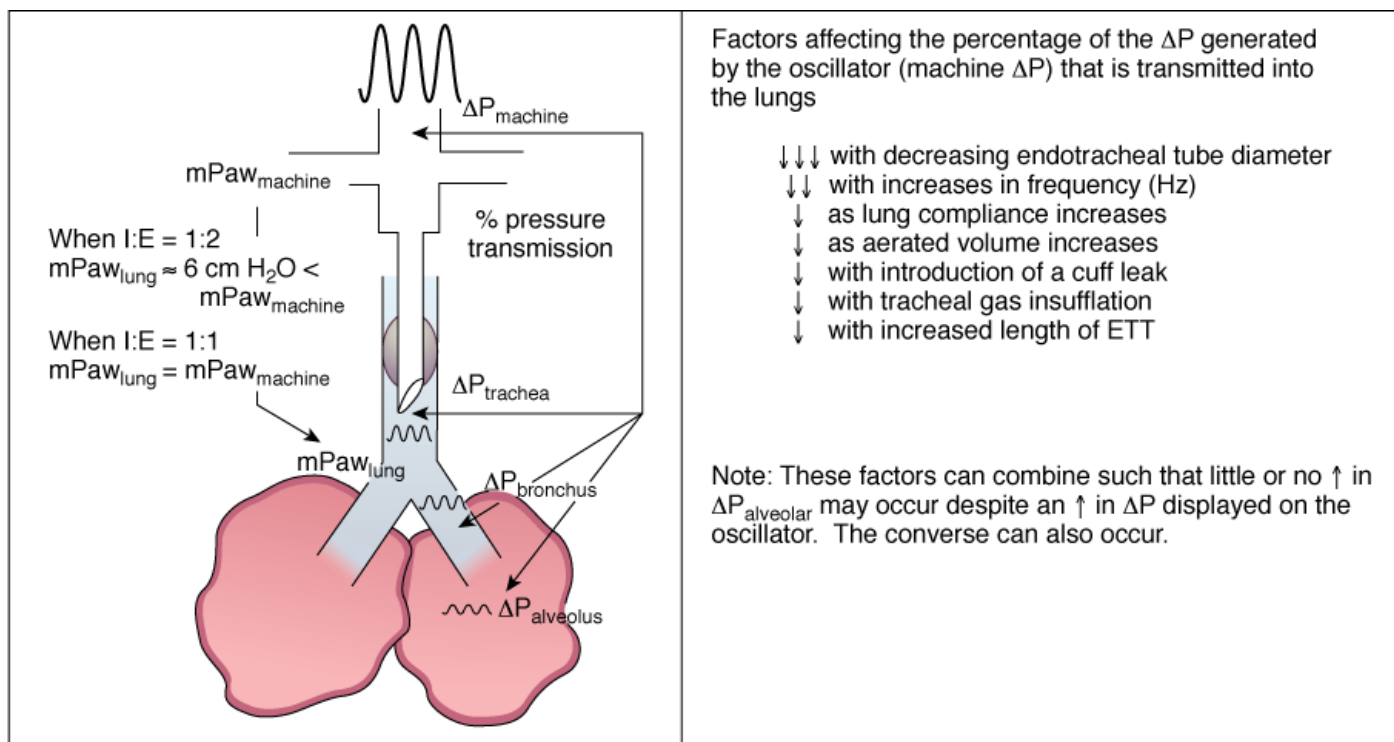
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Schematic of the gas transport pathways during HFOV before and after establishment of a cuff leak. Transport distances are shown for CO_2 . *On the left*, with the endotracheal tube (ETT) cuff occluding the trachea, the fresh-gas front for both CO_2 and O_2 is at the top of the ETT. Gas transport mechanisms must bring CO_2 from the alveoli to the top of the ETT before CO_2 is flushed away by the bias flow. *On the right*, with partial ETT cuff deflation, some of the bias flow passes down the ETT and flushes CO_2 molecules out beside the ETT, in effect reducing the transport distance substantially. The ETT represents 50% of the transport impedance during HFOV.^{71,91} Introduction of a cuff leak reduces the tidal volume needed to support CO_2 elimination and decreases the intrapulmonary ΔP imposed on lung tissue.⁸⁴ A cuff leak also may increase tracheal contamination by oral secretions. The risk-to-benefit ratio of a cuff leak is not yet established.

Frequency Selection

Early HFOV trials in neonates used a frequency of 15 Hz and an inspiration-to-expiration (I:E) ratio of 1:1. Demonstration of small interregional mean pressure gradients at 15 Hz and 50% inspiratory period (T_I)⁹³ subsequently induced a shift to a lower operating frequency of 10 Hz and a 33% T_I in many centers. It is worth noting that although pressure gradients definitely occur in the lung during HFOV, their magnitudes are small (i.e., on the order of 1 to 3 cm H_2O from apex to base).⁹⁴ With a 33% inspiratory period, the mPaw measured in the circuit and displayed on the ventilator exceeds the mPaw within the lung by an amount that increases with tidal volume, frequency, and smaller endotracheal tube diameters, often in the range of 5 to 6 cm H_2O (Fig. 19-8).⁹⁵

Figure 19-8



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Schematic of alterations in mPaw (left side of figure) between the oscillator circuit and lung using two different I:E ratios. Although mPaw values are relatively constant from the machine to the parenchyma with I:E = 1:1, mPaw decreases substantially along the endotracheal tube using an I:E = 1:2.^{73,95} On the right side of the figure the peak-to-trough pressure amplitude (ΔP on the machine) is markedly dampened by passage through the endotracheal tube and airways to the alveoli. Oscillatory waveforms are drawn to scale using data from Sakai.⁷² The associated table indicates multiple factors that reduce ΔP transmission into the lung.

Frequency selection directly affects the pressure cycles applied to the lung. A smaller percentage of the circuit ΔP is transmitted down an endotracheal tube at higher frequencies.^{71,96}

Venegas and Fredberg⁹⁷ approached the question of optimizing frequency during conventional ventilation as a problem of providing adequate alveolar ventilation at minimal pressure cost in a variety of clinical scenarios. Their approach provided graphic depictions of the shifting boundary conditions for safe ventilator settings with varying lung pathology. Although the normal lung can be ventilated over a wide range of frequencies and PEEP levels without inducing dangerous overdistension, in the lung with poor compliance but relatively normal airways PEEP selection becomes critical during conventional ventilation, with penalties in terms of pressure cost and overdistension with either too low or too high a level of PEEP. At high frequencies, the zone of safe PEEP widens, making it easier to maintain more of the lung homogeneously aerated. In lungs with increased airways resistance, the optimal frequency moves to a lower rate. The relationship between frequency and the pressure cost of alveolar ventilation has been developed further by Pillow for HFOV.^{70,71}

Tidal Volume

Tidal volume delivery during HFOV is influenced by many factors (Table 19-1).^{26,98} Adult HFOV initially used frequencies much lower than those used in infants and most animal experiments because machine power appeared inadequate to achieve CO_2 elimination at higher rates. Although the notion that HFOV delivers small tidal volumes has been challenged, newer data in adults suggest that very small tidal volumes can be delivered using adult HFOV, but at very low frequencies, tidal volumes may not be negligible.^{96,99} For optimal lung protection, theoretically one should use the smallest tidal volume and highest frequency that achieves PaCO_2 elimination targets. Fessler et al.¹⁰⁰ studied thirty adult patients on HFOV after failing conventional lung-protective ventilation. CO_2 elimination was adequate in 83% of them using mean maximal frequencies of 9.9 ± 2.1 Hz. Introduction of a cuff leak is recommended before decreasing frequency below 7 Hz.¹⁰¹ Even higher frequencies and small tidal volumes may prove preferable. The relative risk-to-benefit of early cuff leak versus lower frequencies requires

further investigation.^{85,102} Resonant amplification of the delivered volume has been demonstrated,^{103,104} although at a median of 19 Hz in babies, well above rates used clinically.

Table 19-1: Factors Decreasing Delivered Tidal Volume

- ↓ Endotracheal tube diameter (includes secretions, edema fluid)
- ↑ Endotracheal tube length
- ↑ Frequency (most HFOV devices)
- ↓ Power or amplitude
- ↓ Percent inspired time
- ↓ Respiratory system compliance

Although inability to eliminate CO₂ occurred in one of seventeen patients in the original report of Fort et al,⁷⁷ the incidence of ventilation failure (i.e., pH ≤7.15 with bicarbonate ≥19 mEq/L) was the same in both HFOV and conventional ventilation groups in the subsequent larger Multicenter Oscillatory Ventilation for Acute Respiratory Distress Syndrome Trial (MOAT)⁸² and is rare if one follows current recommendations (see “[Troubleshooting](#)” below).

Physiologic Effects of High-Frequency Oscillatory Ventilation

Cardiopulmonary Interactions

Much like high levels of PEEP on conventional ventilation, HFOV is a sensitive detector of intravascular hypovolemia. Patients need intravascular volume repletion before one initiates HFOV and optimizes lung volume. Fortunately, clinical experience and animal studies demonstrate that despite this need for an adequate volume status, treatment with HFOV does not induce “wet lungs.”¹⁰⁵ Alveolar expansion has a powerful impact on pulmonary vascular resistance. Pulmonary vascular resistance goes up both in areas of atelectasis and overdistension. Therefore, the impact of HFOV on pulmonary artery pressure will be influenced strongly by the extent to which homogeneous alveolar expansion can be achieved. This varies with lung pathophysiology, decisions made about mPaw and recruitment techniques, and body position. In clinical experience, pulmonary hypertension secondary to the use of higher mPaw during HFOV has not been a problem. Rather, HFOV often has enhanced the responsiveness to inhaled nitric oxide in both infants and adults.^{66,106–108} As with conventional lung-protective ventilator patterns, patients with preexisting right-ventricular failure need to be watched closely for the possible adverse effects of measures to optimize lung volume on right-ventricular output.

Cardiac output decreases as mPaw increases with any ventilator modality.¹⁰⁹ In early HFOV trials in the premature baboon, cardiac output was less during HFOV⁴⁵ unless mPaw was weaned as lung aeration improved.¹¹⁰ Similar interactions of cardiac output and mPaw occur in infants.¹¹¹ Lesser hemodynamic impact can be expected when ventilating “on the deflation limb,” where alveolar expansion can be maintained at a lower mPaw cost.^{47,52} In adult clinical trials, such as the MOAT⁸² and TOOLS⁵⁴ trials, there were no significant deleterious effects on mean arterial pressure, central venous pressure, pulmonary artery occlusion pressure, or cardiac output compared with conventional ventilation. Small (1 to 2 mm Hg), often transient increases in central venous pressure and pulmonary artery occlusion pressure were observed after transition to HFOV.⁶⁷ Echocardiography has proven useful to assess preload adequacy when clarification is needed.

Interaction with Spontaneous Breathing

Spontaneous breaths contribute forces that can help reexpand alveoli in dependent lung regions¹¹² and improve $\dot{V}_A/\dot{Q}$ distributions.^{113–115} Forel et al,¹¹⁶ however, reported a decreased proinflammatory response in patients given neuromuscular blocking agents for the first 48 hours. A subsequent larger trial found a 28-day mortality difference using lung-protective conventional ventilation with 48 hours of neuromuscular blockade.¹¹⁷ With current HFOV circuit designs (i.e., the lack of an inspiratory demand flow), vigorous respiratory efforts in large patients may cause pressure swings that activate alarms, interrupt oscillations, and produce significant desaturations. For this reason, adults and large children need to have their respiratory efforts suppressed while on HFOV, and initial oscillator trials in adults recommended

routine muscle paralysis. Current protocols attempt to balance the potential pros and cons of neuromuscular blockade in these patients by using paralysis when patients have extremely severe hypoxemia or are generating very large pressure swings despite adequate sedation.^{41,85}

Rationale, Advantages, and Limitations

If we accept that both overdistension and ongoing atelectasis damage lung tissue, then HFOV offers the ability to cycle the largest possible percentage of the total lung within that “safe” range of volume excursion. The key is to decipher exactly when small pressure and volume excursions will make a difference. All therapeutic advances in neonatal and pediatric ventilatory care over the past 30 years that have had a significant impact on survival have exhibited one unifying feature: all have produced more homogeneous lung expansion (Table 19-2).

Table 19-2: Pulmonary Therapies Decreasing Mortality in Neonatal Respiratory Distress Syndrome since 1970

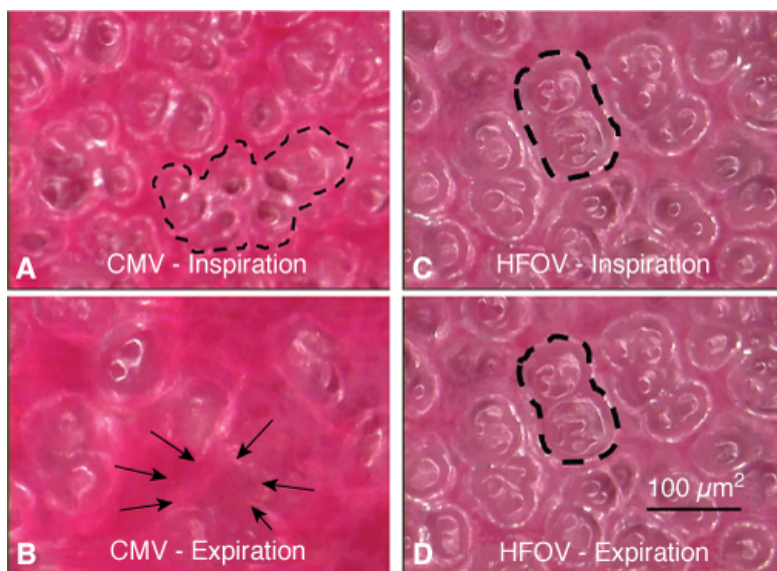
Early use of continuous positive airway pressure
Exogenous surfactant replacement therapy
HFOV/high-frequency jet ventilation with “optimized lung volume”
Conventional ventilation with “optimized lung volume” strategy

Note: All these produced more homogeneous lung aeration.

Advantages of High-Frequency Oscillatory Ventilation

It is much easier in theory than in practice to ventilate the lung within the safe zone of pressure and volumes in *all* lung regions.¹¹⁸ Although much has been written about using the static pressure–volume curve of the respiratory system to guide ventilator settings, any single curve is only a crude composite of the family of pressure–volume curves occurring in different parts of the lungs. A safe peak or plateau pressure will be lower in nondependent and more normal lung regions; closing pressures will be higher in dependent, inflamed, or edematous regions than suggested by a single overall curve.^{7,47,50,55,81} Studies in adults with acute lung injury and acute respiratory distress syndrome demonstrate that derecruitment often begins at PEEP levels substantially above the values recommended as protective of the lung.^{7,119} As one raises PEEP, even acceptable degrees of permissive hypercapnia may be unattainable without exceeding safe peak and plateau pressures at conventional ventilator rates. In such patients, HFOV has the potential to keep that lung in the safe zone with a much larger margin of safety simply because of its ability to sustain gas transport with small volume cycles (Fig. 19-9).³⁸ This is particularly so at higher frequencies because of the impact of time-constant variability on alveolar filling.

Figure 19-9



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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In vivo photomicrographs of subpleural alveoli of a rat after inducing lung injury by saline lavage. Animals were ventilated to equal oxygenation and targets using either conventional mechanical ventilation (CMV) or high-frequency oscillatory ventilation (HFOV). Alveoli in **A** were inflated at end-inspiration during CMV to a peak pressure of 34 cm H₂O but collapsed during expiration **B**, with a PEEP of 9 cm H₂O (arrows). Alveoli were very stable during HFOV at a mean airway pressure of 19 cm H₂O. The same alveolus is seen at inspiration and expiration in panels **C** and **D** (dots). (Reproduced, with permission, from Lippincott Williams & Wilkins, Carney D, DiRocco J, Nieman G. Dynamic alveolar mechanics and ventilator-induced lung injury. *Crit Care Med*. 2005;33:S122–S128.)

A very old but elegant study demonstrated some fundamental features of lung reexpansion risk.¹²⁰ In essence, unless one exceeds airway/alveolar opening pressures, reexpansion will not occur. If one exceeds opening pressure with a long inspiratory period, atelectatic areas open, but normal areas become overdistended. Even very high pressures, however, will not overdistend normal lung if applied very briefly (<0.15 second in this study) because the time constant of highly compliant alveoli is long. Therefore, pressures high enough to exceed opening pressures are safer if applied as brief pulses, as with HFOV. As one turns down the frequency, this safety feature diminishes.

Lung-volume optimization and outcomes equivalent to HFOV have been achieved using recruitment maneuvers followed by rapid conventional ventilation with careful PEEP titration in saline-lavaged animal models.^{121–123} Whether this equivalency extends to more established disease with greater inhomogeneity of opening and closing pressures has not been tested.

Limitations

Most of the perceived limitations of HFOV currently are really limitations of the machines available rather than the technique per se. Even minor modifications trigger expensive Food and Drug Administration (FDA) approval processes. Consequently, oscillator design in North America has been frozen for years.

Increased Use of Muscle Paralysis

The adult-size oscillator currently available in North America does not support concurrent spontaneous ventilation. Infants and children are rarely paralyzed during HFOV. Paralysis of adults on HFOV can be minimized by titrating sedative agents, using higher bias flows to minimize air hunger, and reserving paralytic agents for as-needed bolus use rather than constant infusions.^{41,85,124} Shallow breathing is tolerated in some adults without significant desaturation, particularly during the recovery phase of their disease. Currently, the lack of demand flow to augment spontaneous breaths requires adults to be transitioned to conventional ventilation for final weaning and extubation.⁸⁵

Infection Control

The lack of an expiratory filter initially limited the use of HFOV in patients with infections such as severe acute respiratory syndrome. A suitable filter on the expiratory limb of the circuit is now available (CareFusion, Yorba Linda, CA).

Aerosol Delivery

Metered-dose inhalers are ineffective during HFOV. The Aeroneb nebulizer (Aeroneb Pro, Aerogen, Sunnyvale, CA) delivered the highest percentage of drug in test lung studies.¹²⁵

Transport during High-Frequency Oscillatory Ventilation

It is simplest to execute anticipated transports, such as for computed tomography scan, before starting HFOV, as there is no transport version of HFOV. Many patients receiving HFOV as rescue therapy may be too unstable to transport out of the intensive care unit unless absolutely indicated. Cartotto et al⁶⁶ describe an effective protocol for transport during HFOV, such as to the operating room. It involves clamping the endotracheal tube while on HFOV, transitioning to a self-inflating bag with a 20 cm H₂O PEEP valve, unclamping of the endotracheal tube with vigorous manual ventilation during transport, and reversing these procedures once an oscillator has been set up at the destination. Recruitment maneuvers then are used as needed to reestablish oxygenation at the pretransport HFOV settings.

Monitoring

HFOVs have few ventilator alarms to alert one to a mainstem intubation, tension pneumothorax, or endotracheal tube obstruction. Patterns of possible changes in ventilator readouts with such events have been modeled.⁸⁵ This lack of monitoring on the ventilator makes it critical that clinicians pay attention to both the monitors and the general condition of the patient on HFOV (arterial oxyhemoglobin saturation, heart rate, blood pressure, and so on), and ensure that patients have symmetric chest movements with the oscillations. Absent chest wall movement suggests endotracheal tube obstruction or other ventilator malfunction, while asymmetry may indicate a pneumothorax or mainstem bronchus intubation; sudden increases in ΔP may be seen in both these situations. Also, ventilator noise necessitates brief piston interruption to auscultate heart or breath sounds.

Unfamiliarity of Personnel

A major limitation to the use of HFOV is the level of discomfort of personnel (i.e., physicians, respiratory therapists, and nurses) who encounter it rarely, often as a therapy of last resort.¹²⁶ If one does not understand that a mean pressure of 35 cm H₂O that is stenting 70% to 80% of a stiff lung open while supporting CO₂ elimination with small volume cycles may well induce less lung injury than a PEEP of 15 cm H₂O, plateau pressure of 30 cm H₂O, and larger tidal volumes being distributed to 30% to 40% of that lung, then HFOV “feels” dangerous. Only experience gained in less dire circumstances can produce a comfortable level of understanding of the interaction of ventilator decisions and lung pathophysiology. Then, and only then, can one really test the applicability of any high-frequency modality.¹⁰²

Protocol and Algorithm Limitations

When protocols are generalized to patients outside the population on which the protocol was devised, HFOV may be used inappropriately. Several HFOV protocols specify maximum mean airway pressure limits that were derived in patients with primary pulmonary diffuse airspace disease. If these limits are applied to patients with major extrapulmonary factors, such as abdominal distension or chest wall burn eschar, adequate atelectasis reversal is unlikely because 30% to 70% of the applied mPaw will be lost to inflate the chest wall.⁸¹

Indications and Contraindications

Indications

Is HFOV still needed in the current era of “gentler” conventional ventilation? When HFOV first came into clinical use, large tidal volumes were considered safe during conventional ventilation as long as high peak pressures were avoided.¹¹ In this milieu, randomized, controlled trials in neonates demonstrated substantial outcome advantages from HFOV.¹²⁷ Conventional ventilator protocols then were modified to pursue the same lung protection.

In two recent randomized neonatal studies, significant differences between lung-volume optimizing HFOV and conventional ventilation protocols were seen only in the study in which just 40% of the infants meeting very-low-birth-weight criteria were randomized. These neonates required an FI_{O2} of approximately 0.6 on an mPaw of 8 cm H₂O. In this trial, more neonates managed with HFOV (until extubation) survived

without chronic lung disease and were extubated successfully 1 week earlier than were neonates receiving lung-protective ventilation at conventional frequencies.⁶⁰ When all very-low-birth-weight babies were randomized regardless of O_2 requirements, and on HFOV for only 72 hours, outcome with both ventilators was the same.⁵⁹ This neonatal experience reflects the predictions of the Venegas-Fredberg analysis.⁹⁷ If the lung is fairly normal, a wide range of tidal volumes, PEEP levels, and frequencies can be used without inducing atelectasis or overdistension. Ventilator choice becomes a matter of user preference. As the lung becomes more prone to atelectasis, the range of safe PEEP values shrinks, and injurious extremes of alveolar volume are avoided more reliably at higher frequencies. These are the lungs in which HFOV improves outcome.

Neonatal and Pediatric Patients

Current criteria for using HFOV vary among institutions. Experience in animal models, infants, and children has demonstrated the importance of instituting HFOV early in the lung at risk.^{58,79,80,128,129} Increasingly, extremely low-birthweight neonates are switched to HFOV as their first-intention ventilator in the presence of a rising FiO_2 requirement, $PaCO_2$ greater than 60 mm Hg and/or falling aerated lung volume.^{53,60,130,131} (Courtney SE, personal communication.) A postrecruitment FiO_2 of less than 0.30 is taken as evidence of adequate alveolar reexpansion following early institution of HFOV.^{62,130}

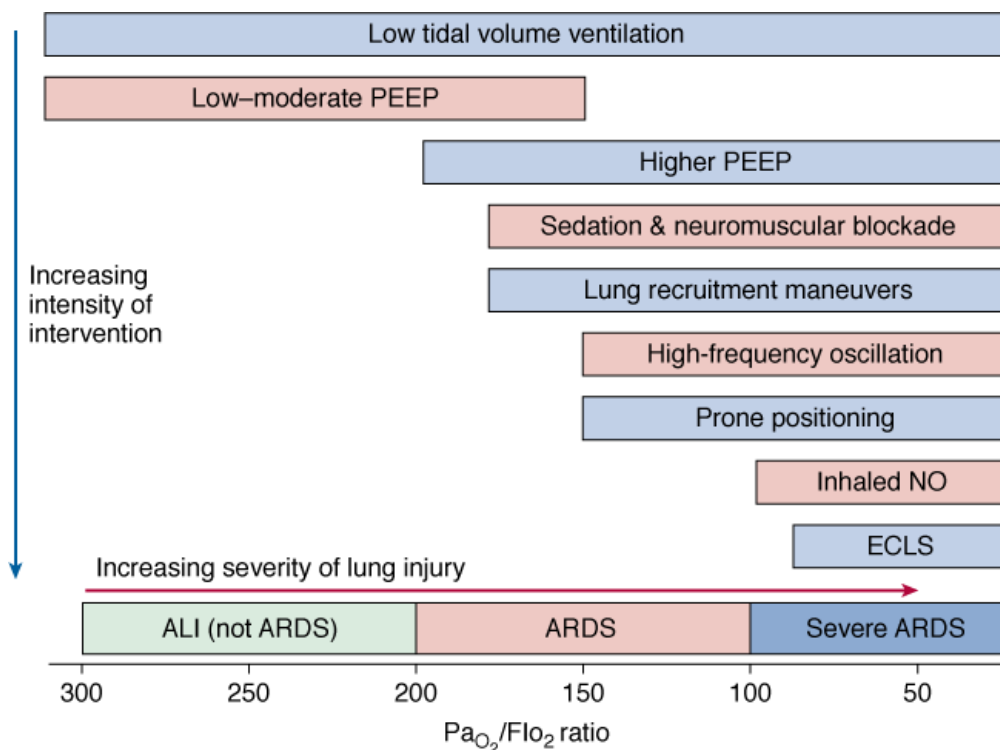
Tissieres et al¹³¹ found that 41.7% of neonates managed with early lung recruitment on HFOV achieved adequate oxygenation without exogenous surfactant, required shorter periods of ventilator support, and had a lower incidence of hemodynamically significant patent ductus. In pediatric patients, Duval et al¹²⁶ switched to HFOV at an oxygenation index greater than 13 or a pH less than 7.25 despite a bicarbonate level of 20 mmol/L or greater and peak pressures of more than 30 cm H_2O . In older children, Doctor and Arnold¹³² recommend institution of HFOV if an “open-lung strategy” using conventional ventilation generates a peak pressure of greater than 35 cm H_2O despite permissive hypercapnia or $mPaw$ values approach 15 to 18 cm H_2O and the still exceeds 0.6.

Pulmonary hypertension occurs frequently in neonates. It can generally be reversed using HFOV (or high-frequency jet ventilation) to achieve better CO_2 elimination and normalize pH values. Randomized, prospective studies indicate that when inhaled nitric oxide is needed, therapeutic response is improved with concurrent institution of HFOV in both neonates and children.^{108,133,134} All these guidelines represent an attempt to define the characteristics of a lung for which HFOV might be more protective than a conventional lung-protective strategy.

Adult Patients

Indications for HFOV are evolving in adult lung disease. These generally fall into two categories: first, use of HFOV when the patient is failing conventional therapy, either because of frank desaturations on high FiO_2 or the requirement of high airway pressures to achieve adequate gas exchange (Fig. 19-10). Second, use of HFOV for the primary prevention of ventilator-induced lung injury, in patients with severe acute respiratory distress syndrome. Although it may not be an independent predictor of mortality,¹³⁵ a consistent theme for both indications is that survival is better with early (<3 days) rather than late (>7 days) institution of HFOV.^{65,77}

Figure 19-10



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Schematic of various therapies for acute lung injury and acute respiratory distress syndrome and their applicability across the spectrum of disease severity. Some interventions (e.g., low tidal-volume ventilation) are broadly applicable, whereas others, such as extracorporeal life support (ECLS) and inhaled nitric oxide (iNO), should be reserved for the most severely hypoxemic patients.^{16,17}

Clinicians typically turn to HFOV as a potential therapy^{77,78,136} in the difficult-to-oxygenate patient when FiO_2 is above 0.8 despite a trial of high PEEP. In this setting, data from observational studies and randomized controlled trials suggest that HFOV is safe and effective in improving oxygenation.^{85,101,102} Whether HFOV is more effective than lung-protective conventional ventilation at preventing ventilator-induced lung injury and reducing mortality remains a question for debate. The largest randomized controlled trial comparing HFOV with conventional ventilation in adults published to date ($N = 148$) found a trend toward improved mortality with HFOV (absolute risk difference: 0.15; $p = 0.1$).⁸² The tidal volumes used in the control arm of this trial exceeded current recommendations. This may have contributed to excess mortality in the control arm.

Recently, all of the randomized controlled trials comparing HFOV to CV have been summarized in a metaanalysis (eight trials; $N = 419$).¹⁵ Although none of the trials individually showed a mortality benefit, the combined results suggest a statistically significant reduction in mortality (relative risk [RR] 0.77; 95% confidence interval [CI] 0.61 to 0.98; $p = 0.03$). We view this as a hypothesis-generating result, given that it is a meta-analysis of several small trials, many with methodologic problems.

In experienced centers, a trial of HFOV may be considered when the FiO_2 requirements remain greater than 0.6, the plateau pressure approaches 30 cm H_2O , and PEEP is 10 to 15 cm H_2O .^{85,101,102} The current challenge is to define the pulmonary characteristics that indicate that the lung is at risk of ventilator-induced lung injury using conventional ventilation and then to test whether HFOV is of benefit in that population. The goal is to identify a group of patients with a significant inflammatory lung injury who may benefit from lung recruitment. Indicators such as PEEP responsiveness,¹³⁷ dead-space measurements,¹³⁸ and biomarkers¹³⁹ are being explored. Until additional information becomes available, indications are that it is patients with the most severe lung injury who are likely to be the most recruitable and most at risk for ongoing ventilator-induced lung injury with lung-protective conventional ventilation.^{6,76} Indicators ideally would include an accurate bedside method to determine the extent of inhomogeneity of alveolar expansion across lung regions, both at end inspiration and at end expiration.^{47,81,87}

Contraindications

Early literature advised against the use of HFOV in obstructive airway diseases, such as asthma, because of the theoretical risk of inadvertent gas trapping and hyperinflation. Recent reports demonstrate that HFOV can be used in such patients, even in patients demonstrating gas trapping on conventional settings. Case reports of patients with severe asthma and a series of infants with respiratory syncytial virus infection with severe bronchiolitis describe what is being termed the *open-airway strategy* for the management of such patients.^{36,140} The goal in these patients is to find the narrow window of mean airway pressure that stents the diseased airways open enough to support high-frequency gas transport without inducing excessive lung distension with hemodynamic impairment. Extensive experience with HFOV is recommended before one pursues this application.³⁶ HFOV remains contraindicated in localized airway obstruction or bullous disease.

Before development of an expiratory gas filter, severe acute respiratory syndrome and other highly contagious airborne pathogens were a relative contraindication to HFOV.¹⁴¹

Comparison with Other Modes

Limited comparisons are available among various high-frequency modalities. When high-frequency jet ventilation was introduced into adult and neonatal use, the primary goal was to support gas exchange at the lowest possible peak and mean airway pressures to facilitate resolution of barotrauma. When HFOV was introduced in an early neonatal clinical trial, the primary goal was to optimize the volume of aerated lung. Both devices, in fact, can be used with either a low-pressure or open-lung strategy.⁵⁸ The strategy must be matched to the patient's pathophysiology. Only one study systematically compared three high-frequency options in an animal model of the atelectasis-prone lung.¹⁴² Optimal outcome occurred with the mode (HFOV) in which the end-expiratory pressure was closest to the mean pressure level and the least time was spent at the end-expiratory level.

Variations in Delivery

The performance of several commercially available devices for delivery of HFOV to neonates has been evaluated both in bench studies and in animal models. Substantial differences in performance have been demonstrated.^{73,74,143} Any ventilator introduced into an intensive care unit should be thoroughly understood before clinical use. For example, all neonatal ventilators tested by Hatcher et al⁷³ and Pillow et al⁷⁴ showed a decrease in delivered volume with a decrease in lung compliance, but one ventilator (no longer available) exhibited the reverse phenomenon.¹⁴³ That outlier characteristic may explain puzzling episodes of hypercapnia during the weaning of infants with that ventilator. One ventilator failed to increase delivered volume at settings between 50% and 100% of its amplitude dial. It also was the only ventilator that markedly increased tidal volume as mean pressure increased.^{73,143} Ventilators varied in the relationship of displayed mPaw and intrapulmonary mean pressure measured during endotracheal tube occlusion.⁷³ Threefold differences were documented in maximum tidal volume among ventilators tested at 15 Hz,^{73,74} and frequency dependence of tidal volume varied substantially. It is amazing that HFOV has been used as safely and effectively as it has considering these major discrepancies in ventilator performance among the devices used.

At the time of writing, only one oscillator is approved in the United States and Canada (it is also available in Europe) for use in adults and large children (SensorMedics 3100B, CareFusion, Loma Linda, CA). Another device is approved in Europe and Japan, marketed as the Vision α (Novalung, Baden-Württemberg, Germany) or the R100 (Metran Corporation, Tokyo, Japan). In contrast to the 3100B, the Vision α also provides the ability to deliver several conventional modes of ventilation with the same circuit. The two devices also differ in how they generate and modulate the oscillatory waveforms during HFOV. We are not aware of any published studies comparing the technical or clinical performance of these two oscillators.

Adjustments at the Bedside

Preparation for Initiation of High-Frequency Oscillatory Ventilation

Table 19-3 lists suggested strategies for management of HFO in adults with recent-onset acute lung injury and acute respiratory distress syndrome.

Table 19-3: Suggested Strategies for Management of High-Frequency Oscillation in Adults with Recent-Onset Acute Lung Injury and Acute Respiratory Distress Syndrome

	Titration Using a mPaw/ FiO_2 Table	Alveolar Reexpansion Given Priority
Initial Settings	FiO_2 1.0; mPaw 30 to 35 cm H ₂ O; I:E 33%; amplitude (ΔP) = 90 cm H ₂ O; frequency based on recent pH	
O₂ Management	- Initial RM (40 cm H₂O × 40 s) ± repeat RM - Use mPaw/FiO_2 table to decrease both concurrently - Move up or down mPaw/FiO_2 table as needed according to SpO_2 - Repeat RM before increases in mPaw to > 25 cm H₂O - Repeat RM for hypoxemia with disconnects etc.	- Initial RM (40 cm H₂O × 40 s) ± repeat RM - Increase RM pressure and mPaw as needed to decrease required FiO_2 balancing pressure cost vs. recruitability - Hold mPaw that maintains oxygenation after RM - Wean FiO_2 until = 0.4 - Wean mPaw incrementally as long as FiO_2 stable - Repeat RM before any increases in mPaw - Repeat RM for hypoxemia with disconnects etc.
CO₂ Management	- Target pH 7.25 to 7.35 using highest frequency (f) possible; maintain ΔP = 90 cm H₂O - Consider a partial endotracheal tube cuff leak if $f \leq 6$ Hz	
Transition to CV	- Convert when FiO_2 = 0.4, mPaw = 20 to 24 cm H₂O and stable > 12 hours - Initial settings: V_T 6 mL/kg PBW; FiO_2 0.5; PEEP 16; RR 25	
Return to HFO	- Return to HFO if required $\text{FiO}_2 \geq 0.6$ to 0.7 on PEEP ≥ 14 - Return to HFO if pH < 7.25 with V_T 6 mL/kg PBW and $P_{\text{PLAT}} > 30$ cm H₂O	- Return to HFO if $\text{FiO}_2 \geq 0.5$ on above settings - Return to HFO if pH < 7.25 on above settings

Abbreviations: CV, conventional ventilation; ΔP , oscillatory pressure amplitude; FiO_2 , fractional inspired oxygen concentration; HFO, high-frequency oscillation; I:E, inspiratory-to-expiratory ratio; mPaw, mean airway pressure; PBW, predicted body weight; PEEP, positive end-expiratory pressure; P_{PLAT} , inspiratory pressure plateau; RM, recruitment maneuver; SpO_2 , oxygen saturation; V_T , tidal volume.

Note: This table outlines an overall approach while indicating areas in which different experts use somewhat different decision algorithms.

Source: Modified, with permission, from Fessler HE et al.¹⁰¹

Documentation of a patent endotracheal or tracheostomy tube is essential.⁴¹ Partial obstructions impede gas transport much more during HFOV than with conventional modes. Tubes that allow passage of a suction catheter still may have a narrow lumen. If the patient has been intubated for more than 2 or 3 days, a preemptive bronchoscopy may be useful to verify tube patency as well as provide opportunity for removal of any mucus plugs. Hypovolemic patients will not tolerate the high mPaw used with HFOV; adequate volume status should be ensured before initiating HFOV.

Sedative and analgesic drugs should be titrated while the patient is still receiving conventional ventilation. Although muscle paralysis was used in most adult published trials, subsequent experience has found continuous paralytic infusions unnecessary in many cases when appropriate levels of sedation are established.^{41,101} Small as-needed doses of muscle relaxants may facilitate the initial period of adjustment to HFOV and may be needed intermittently during periods of agitation to facilitate retitration of sedative and analgesic agents.^{85,124} Preservation of some spontaneous respiratory effort is valuable, as with any type of prolonged ventilation, provided those efforts do not produce marked fluctuations in mPaw (>5 cm H₂O) or O₂ saturation (>3% to 5%). Careful observation for spontaneous breathing following

transition to HFOV allows proper adjustment of high-pressure and low-pressure alarms. Patient comfort may be enhanced when bias flows of 40 to 60 L/min are used to achieve the target mPaw in adults.

Oxygenation

The approach used in most early algorithms and trials initiated HFOV with an FiO_2 of 1, an mPaw 5 cm H_2O higher than the mean pressure on conventional ventilation, and a 33% inspiratory time. Inability to reduce the FiO_2 below 0.6 was managed primarily with stepwise increases in mPaw, in essence gradually moving the respiratory system up the inflation limb of the pressure–volume relationship. More recent protocols apply one or more lung-recruitment maneuvers when HFOV is initiated, along with rapid upward titration of mPaw, with the goal of optimizing alveolar reexpansion to place the lung on the deflation limb.^{54,101} Starting mPaw levels of 30 to 35 cm H_2O are now recommended in moderate to severe acute respiratory distress syndrome. Table 19-3 highlights two current approaches to optimizing oxygenation during HFOV.¹⁰¹ Both utilize initial recruitment maneuver(s) but then differ on whether both mPaw and FiO_2 should be reduced in tandem (using a table) or whether measures to reduce FiO_2 should take priority over mPaw reduction (with FiO_2 being used as a surrogate measure of alveolar reexpansion).¹⁰¹

Recruitment maneuvers are performed by resetting the high-pressure alarm to a higher value (e.g. 50 cm H_2O), eliminating a cuff leak if present, turning off the piston, raising the mPaw slowly to 40 to 45 cm H_2O (maximum 50 cm H_2O) for 40 to 60 seconds, returning the mPaw to the original setting (or 3 cm H_2O higher), and then restarting the piston and restoring cuff leak and alarms to the original levels. Recruitment maneuvers should be considered if desaturation occurs after suctioning, bronchoscopy, circuit disconnects, or patient repositioning.

In the TOOLS trial, one to three recruitment maneuvers were performed immediately on initiating HFOV.⁵⁴ This methodology matches protocols used in many animal studies of lung protection using HFOV. With this protocol, an FiO_2 of less than 0.6 was achieved in 68% of patients by the end of the initial recruitment cycle, which delivered a mean of 2.4 recruitment maneuvers over 1.5 hours.

With both approaches, the accepted target has been projection of the diaphragm onto the eighth or ninth rib posteriorly on an anteroposterior film. An alternative reference in adults is visualization of the fifth rib above the diaphragm anteriorly or an apical diaphragm distance not greater than 25 cm.¹⁴⁴ Emerging techniques for the evaluation of regional expansion hopefully will soon replace these crude measures.⁸⁷ Information on the relative timing, efficacy in terms of achieving the most homogeneous lung expansion pattern, and complication rate of these two approaches is needed before it will be possible to define an “optimal” approach. The potential hazard of the first approach is too little recruitment too slowly. The potential hazard of the second approach is excessive lung distension at a rote value of 30 cm H_2O mPaw if that value is not reduced expeditiously enough in a lung that is very responsive to recruitment. What is becoming clear is the importance of ventilating “on the deflation limb,” whether with conventional ventilation or with HFOV.^{52,145}

It remains speculative whether the target FiO_2 for optimal lung protection should be 0.4 or 0.6. A substantial amount of ongoing atelectasis remains at an FiO_2 of 0.6. As one moves toward an earlier transition to HFOV—as in current neonatal and pediatric experience in many centers—lower FiO_2 targets become achievable at “reasonable” mean pressures. At high FiO_2 requirements, mPaw levels are reduced only when doing so does not increase FiO_2 requirement. When the FiO_2 is stable at 0.6, some protocols recommend reduction of mPaw before further FiO_2 decreases if the mPaw is greater than 35 cm H_2O . Whether this compromise between ongoing atelectasis and the risk of higher pressures is optimal will likely only be resolved when bedside methods of assessing areas of overdistension become available. One protocol that recommends this mPaw/ FiO_2 compromise also recommends placing the patient in a prone position if the FiO_2 requirement remains 0.6 or more in order to potentiate alveolar recruitment in dorsal lung regions.⁴¹

Carbon Dioxide Elimination

The optimal approach to CO_2 elimination remains controversial. When should the transport advantages of an endotracheal tube cuff leak be introduced? Inasmuch as the lung-protective potential of HFOV lies in its small volume cycles, it would seem logical to introduce it at the time of initiating HFOV in a large patient. With a cuff leak of 5 to 8 cm H_2O , one can eliminate CO_2 at a higher frequency, deliver the smallest possible tidal volumes, and produce the lowest possible ratio of intratracheal to airway opening pressure swings.^{84,96} With this approach,

even very heavy patients generally can be ventilated at frequencies of 6 Hz or more. If reductions in frequency are required at any stage, the first response to a decrease in PaCO_2 with recovery should be an increase in frequency. Whether the frequency target in adults should be 8 Hz or higher is presently unknown. Detailed protocols differing in their relative weighting of amplitude (ΔP) and frequency are available.^{85,101,102}

An alternative approach is progressive reduction of frequency to 3 Hz as needed to achieve target CO_2 levels, with a cuff leak being added only when these much larger stroke volumes prove inadequate.

The advantage of instituting a cuff leak when initiating HFOV is the ability to use higher frequencies and smaller volume cycles. The disadvantages are a need for more nursing care to ensure removal of oral secretions and respiratory therapy involvement to remove and then reset the cuff leak whenever a recruitment maneuver is performed. Endotracheal tubes with infraglottic suction ports may prove useful in this context. In cases of severe upper airway swelling, a pharyngeal airway positioned just above the larynx can be used to provide an exit pathway for a cuff leak, as described by Cartotto et al⁶⁶ in a burn patient. Failure of mPaw to drop when instituting a cuff leak indicates swelling around the cuff or upper airway.

Protocols also differ in their approach to setting power or ΔP . Some advocate starting at maximal power so that one can use the highest frequency possible;¹⁰¹ other algorithms manipulate both ΔP and frequency settings.^{41,85} Current protocols favor targeting a relatively high and constant ΔP to enable one to use the highest possible frequency to deliver the lowest tidal volume and smallest intrapulmonary pressure swings that can achieve adequate CO_2 elimination. Transcutaneous PaCO_2 sensors can provide useful trend information (not absolute values) to guide frequency adjustments during the initiation phase if rapid point-of-care blood-gas analysis is unavailable. Tighter PaCO_2 targets are needed in patients with elevated intracranial pressure, as well as in neonates in whom the risk of retinopathy of prematurity may increase at high PaCO_2 ⁸ while low PaCO_2 levels produce neurodevelopmental deficits.^{146,147}

Patient Positioning

Patients should be placed at 30 degrees head up unless hemodynamically unstable. The prone position also can be used during HFOV to optimize expansion of dorsal lung regions, with appropriate protocols for caring for the prone patient.¹⁴⁸ If the prone position is not feasible but hypoxemia remains severe, switching the patient from side to side in true lateral positions (i.e., 90 degrees to the mattress) may sometimes improve lung recruitability.

Troubleshooting

Partial Pressure of Arterial Carbon Dioxide Problems

It is particularly important to troubleshoot PaCO_2 problems appropriately (Table 19-4).

Table 19-4: PaCO₂ Problems

PaCO ₂ Too Low	PaCO ₂ Too High	Acute Increase in PaCO ₂
↑ f in increments of 1 to 2 Hz to 8 ⁹⁰ or 12 Hz ⁸⁹	↑ ΔP if not already at 90 cm H ₂ O in increments of 5 cm H ₂ O	Verify ETT is patent: • Pass suction catheter • Quick bronchoscopy • Can one ventilate adequately with manual bag/mask?
↓ ΔP in 5 cm H ₂ O increments	Institute a cuff leak if not present	
	Try RM(s) to reverse atelectasis	Is there a pneumothorax?
	Is a higher maintenance mPaw needed? (CXR or other measure of lung volume) ↓ f to a minimum of 3 Hz in 1 to 2 Hz decrements	• Decreased vibrations unilaterally? • CXR Progressive Increase PaCO₂ Is chest wall compliance decreasing? • Abdominal compartment syndrome • Burn eschar

Abbreviations: CXR, chest X-ray; ΔP, oscillatory pressure amplitude; ETT, endotracheal tube; f, respiratory frequency; RM, recruitment maneuver.

Decreased Partial Pressure of Arterial Carbon Dioxide

As PaCO₂ levels decrease during the course of lung recovery, the first response should be to increase frequency back up to the 8- to 10-Hz range (1- to 2-Hz increments) and then to decrease power (5 to 10 cm H₂O ΔP decrements) in order to minimize volume and pressure swings in the lung. A decrease in ΔP at a constant power setting may be an indicator that lung volume is increasing.

Increased Partial Pressure of Arterial Carbon Dioxide

An abrupt substantial increase in PaCO₂ in a previously stable patient may reflect plugging of the endotracheal tube, development of a pneumothorax, or atelectasis. One should ensure passage of a suction catheter, check one's ability to manually bag or mask ventilate, and perform a quick bronchoscopy while waiting for a chest X-ray to clarify the etiology. Before decreasing frequency or increasing power in response to a gradual increase in PaCO₂, one must ensure that the lung is expanded adequately. An abdominal compartment syndrome should be considered. Available oscillators all decrease delivered stroke volume when faced with a greater load.^{73,74} Appropriate recruitment procedures often can resolve PaCO₂ problems, while at the same time decreasing the phasic pressure swings within the airways and alveoli. A cuff leak should be considered before lowering frequency.

Oxygenation Problems

As discussed earlier, oxygenation depends on achieving and then maintaining end-expiratory lung volume. A gradual downward drift of O₂ saturation after a recruitment maneuver is indicative of too low a maintenance mPaw. Recruitment should be repeated with a return to an mPaw 2 to 3 cm H₂O higher than the previous level. If the FI_{O2} requirement remains greater than 0.60 despite recruitment maneuvers, prone positioning should be considered.¹⁶ Inhaled nitric oxide can be added, although whether this improves outcome as well as oxygenation remains unknown.¹⁰⁸ In the severely hypoxemic patient, HFOV is one of a number of recommended techniques (see Fig. 19-10).^{16,17}

Air Leaks

Barotrauma occurs at similar frequencies during HFOV as with conventional ventilation, but may be more subtle in its manifestations. Displayed mPaw will remain stable even with a tension pneumothorax. The gradual development of hypotension and desaturation, with or without a unilateral decrease in chest wall motion, should trigger a portable chest X-ray. In an experimental model of pneumothorax during HFOV, air leak was minimized by the use of higher frequency, lower ΔP , lower mPaw, and shorter inspiratory period.¹⁴⁹ Excessive decreases in mPaw will promote atelectasis with a resulting increase in intrapulmonary pressure cost, delaying resolution of the air leak. In general, recruitment maneuvers are not advised in the presence of air leak, but this is not an absolute contraindication. In the presence of severe hypoxemia and unilateral air leak, recruitment maneuvers performed with the patient in a full lateral position with the leak side down may achieve significant benefit by reexpansion of the nondependent lung.

Hemodynamic Compromise

Inability to tolerate the institution of HFOV or application of recruitment maneuvers may reflect an inadequate intravascular volume. Duval et al¹⁴⁰ reported a need for a transient increase in fluids and inotropic agents in some infants during transition to HFOV. Interpretation of filling pressures (central venous pressure, pulmonary artery occlusion pressure) may be difficult in the setting of high mPaw. A fluid challenge, cardiac echo study, or other assessment of cardiovascular status may be needed in ambiguous situations.⁴¹

Important Unknowns

HFOV protocols are under constant revision through animal experiments and clinical trials that evaluate both the gas exchange and ventilator-induced lung injury impact of ventilator setting decisions. We need criteria that indicate when any given ventilator pattern needs to be replaced by one with a greater potential for lung protection. This is a complex question considering that we do not even know when permissive hypercarbia becomes deleterious for patients.

The optimal frequency or frequencies for minimizing the intrapulmonary pressure and distension “cost” of HFOV in the adult lung with varying pathophysiologies remain unknown. Such information would clarify the risk-to-benefit ratio of early use of a cuff leak. Existing algorithms differ substantially in their management of frequency.^{85,101}

Optimal methods of aerosol delivery during HFOV need further evaluation. Better bedside knowledge of which techniques (e.g., recruitment maneuvers, ventilator settings, position changes [prone/supine], and so on) produce the most homogeneous lung expansion across all lung regions would greatly aid decision making. Emerging technologies, such as electrical impedance tomography, show promise.

Neonates often are maintained on HFOV until extubation. Premature transition to conventional ventilation can negate the benefit of HFOV. Currently spontaneous breaths are not augmented during HFOV, so this approach is not feasible in large patients. This may change if demand flow is incorporated into new versions of this device.¹⁵⁰ The “optimal” timing of transition to conventional ventilation may be difficult to determine until this device limitation is resolved. It is well recognized that gross deterioration of oxygenation after transition to conventional ventilation should trigger an immediate reevaluation of the patient and possible return to HFOV.

Future Directions

Two large multicenter, randomized, controlled trials are currently underway comparing HFOV to conventional ventilation: OSCILLATE (ISRCTN87124254) and OSCAR (ISRCTN10416500). The OSCILLATE trial (target N = 1200) attempts to optimize lung volume “on the deflation limb” of the pressure–volume relationship and is recruiting patients from centers with significant experience with HFOV. The OSCAR trial (target N = 1006) employs a more traditional stepwise increase in mPaw (walking up the inflation limb of the pressure–volume curve), and is also assessing the applicability of HFOV as a novel technology in inexperienced centers. Currently, many centers use HFOV rarely and only for severe, refractory oxygenation failure or air-leak syndromes associated with end-stage lung disease. This is not the way to develop a cadre of physicians, nurses, and respiratory therapists who can skillfully troubleshoot interactions between the machinery and the patient’s pathophysiology.

Redesigned Machines

Currently, high-frequency oscillators are still classified as class 3 devices by the FDA. This means that any redesign triggers expensive review costs, much in excess of that required for bringing a new conventional ventilator to market. Revision of this classification would greatly

facilitate improvements in device design.

Clinical Lung-Volume Monitoring

Developments in this area should take a lot of guesswork out of lung-volume optimization during both conventional and high-frequency ventilation. Hopefully, future guidelines for the initiation of HFOV will be able to include some measure of interregional inhomogeneity of alveolar expansion as one of the criteria, not just a plateau pressure, FI_{O_2} , or pH value.

Summary and Conclusion

High-frequency ventilators expand our ability to minimize both atelectrauma and overdistension injury in the vulnerable lung. Evidence of the benefits of early institution rather than late “rescue” use is growing. Optimal use requires constant attention to the interplay between the patient’s pathophysiology and machine factors to keep the lung in its “safe zone.” The full potential of HFOV will only be clarified when well-controlled randomized clinical trials are done that incorporate both early intervention and lung recruitment protocols that rapidly place the lung on the deflation limb of its pressure–volume relationship. Such trials need to compare HFOV to the lung-protective conventional ventilation protocol considered optimal at the time of study.

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